

ODER: Origins of Descriptive Representation

Political Economy Workshop @ BI

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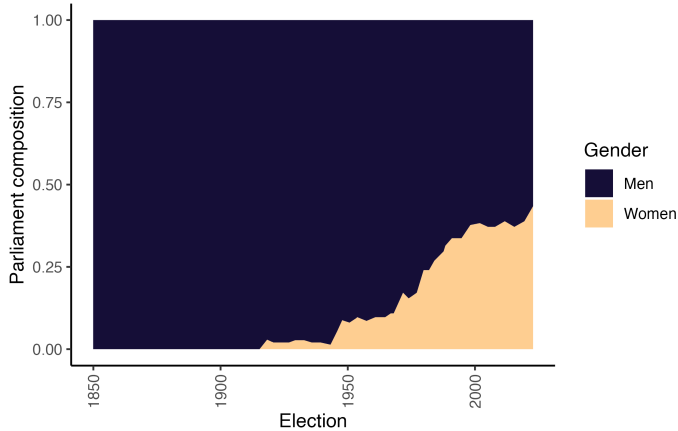
Legislatures don't resemble voter populations



Descriptive underrepresentation of women and workers



In Denmark, slow march to ~40 pct. women



Klint, T., Evert, A. S., Kjær, U., Pedersen, M. N. & Hjorth, F. (2023). The Danish Legislators Database. *Electoral Studies*

Schwarz & Coppock (2022): Voters prefer female candidates

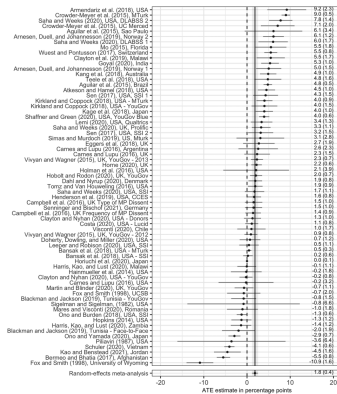
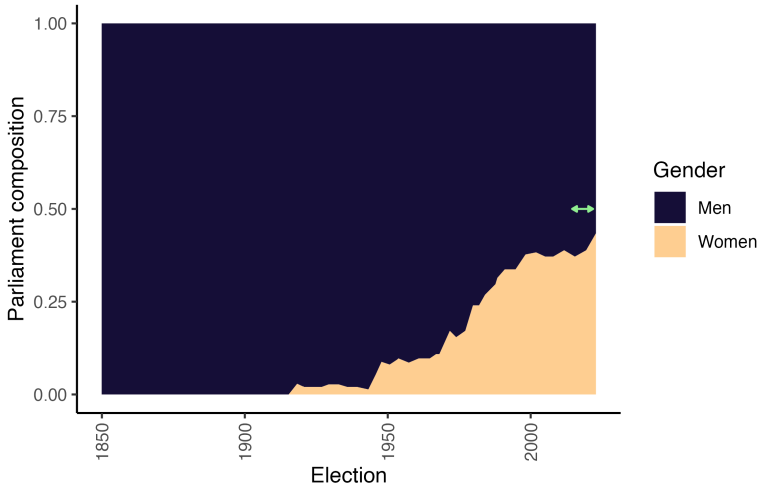


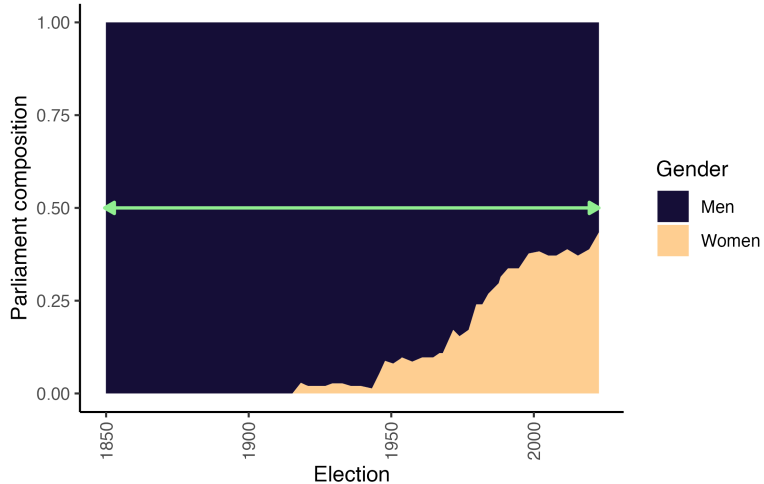
Figure 1. Results of 67 candidate choice experiments on the effect of candidate gender

Schwarz, S., & Coppock, A. (2022). What have we learned about gender from candidate choice experiments? A meta-analysis of sixty-seven factorial survey experiments. *The Journal of Politics*, 84(2), 655-668.

But: based only on studies 2014-



What if we considered the entire range?



- with historical scope, we can examine the original obstacles to descriptive representation
- *breakthrough candidacies*: the first-ever election of a candidate from an underrepresented group in a given electoral unit

RQ

How do breakthrough candidacies affect electoral support, nominations, and rhetoric among candidate parties and their competitors?

↪ can enrich understanding of obstacles for currently unrepresented groups

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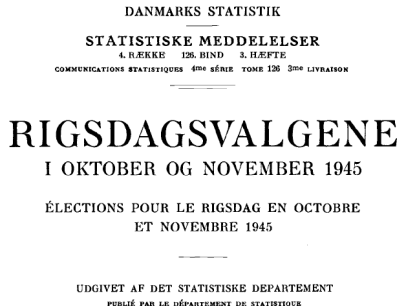
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Data source: historical Danish election reports



Example: Social Democrats, Copenhagen, 1945

Vestre Storkreds i Hovedstaden							
	Kbh. 1. Kreds	Kbh. 13. Kreds	Kbh. 14. Kreds	Frb. 1. Kreds	Frb. 2. Kreds	Frb. 3. Kreds	Til- sammen
A. Socialdemokratiet:							
1. Alsing Andersen	518	7 805	1 814	337	432	661	11 567
2. J. Chr. Jensen	4 258	279	280	56	95	92	5 060
3. Fr. Dalgaard	88	139	15 256	61	90	78	15 712
4. Einer-Jensen	148	180	648	284	8 397	633	10 290
5. Fanny Jensen	360	383	1 075	352	433	407	3 010
6. Oluf Bertolt.....	13	29	71	14	64	3 206	3 397
7. Edvin Dose	28	73	202	1 677	96	39	2 115
Tilsammen...	5 413	8 888	19 346	2 781	9 607	5 116	51 151
heraf personlige Stemmer.....	4 156	7 375	14 439	1 908	7 965	3 704	39 547

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Goal: candidate-level vote data

Name	Dist1	Dist2	Votes
⋮	⋮	⋮	⋮
Fanny Jensen	Vestre Storkreds	Kbh. 1. Kreds	30
Fanny Jensen	Vestre Storkreds	Kbh. 13. Kreds	383
Fanny Jensen	Vestre Storkreds	Kbh. 14. Kreds	1075
Fanny Jensen	Vestre Storkreds	Frb. 1. Kreds	352
Fanny Jensen	Vestre Storkreds	Frb. 2. Kreds	433
Fanny Jensen	Vestre Storkreds	Frb. 3. Kreds	407
⋮	⋮	⋮	⋮

How many vote counts? A back-of-envelope estimate

$$\left(\underbrace{6 \times 8}_{\text{counts per table}} \times \underbrace{4 \times 32}_{\text{tables per report}} \right) \times \underbrace{60}_{\text{elections}} = 368.640$$

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Solution: Use computer vision to automate digitization

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Inferring candidate characteristics from names + occupations

II. Vestre Storkreds (Partiliste; Fordelingstal: 12 788):

1. Fhv. Minister Alsing Andersen*, Kbh.....
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3. LRSagfører Fr. Dalgaard*, Kbh.....
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↪ use word embeddings to scale class alignment of occupation labels

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↪ use word embeddings to scale class alignment of occupation labels

Pilot: Occupation class alignment from word embeddings

- pretrained Word2Vec embeddings
- based on 30M Danish newspaper pages 1880–2005
- define ‘workerness vector’ $\vec{W} = \vec{\text{worker}} - \vec{\text{capitalist}}$
- calculate occupations’ cosine similarity with $\vec{W}$

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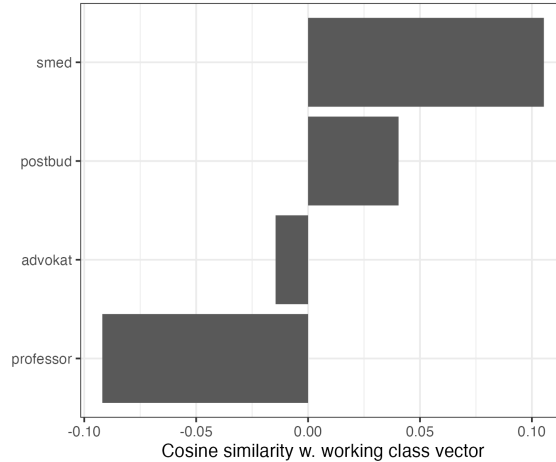
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Results for select occupations



Observing effects on rhetoric using digitized parliamentary debates

115 ²⁷/₁₁ 47: 1. Beh. af F. t. L. om Ændr. af L. om Lukketid for Butikker og Lagre m.v. 116

[N. Chr. Christensen.]

ficeret Arbejdskraft, naar man kan byde paa den i vor Tid saa eftertragtede Frihed —, har Medaillen alligevel sin Bagside; som kan give Anledning til Betænkeligheder, fordi Problemet nu engang ikke er saa ligetil, som man vil give det Udseende af.

Spørgsmaalet bliver derfor, om ikke de nugældende Fordele er for store, om ikke denne Lukkelov i et for urimeligt Omfang gaar ud over Forbrugerne, fordi Friheden er gaaet ud over det rimeliges Grænse, og om ikke Ordningen ogsaa national-økonomisk har gjort Skade ved, at Produktionen gaar ned, Indtægterne formindskes,

lever man i mange Industrier, hvor der er beskæftiget kvindelige Arbejdere, at man har svært ved at faa den kvindelige Arbejdskraft frem om Lørdagen. Der er paa denne Dag en meget stor Forsømmelsesprocent, og det bidrager til at nedsætte Produktionen. Hvis jeg ikke husker meget fejl, har det nuværende højtærede Medlem af Regeringen, Fru Fanny Jensen, ved en Lejlighed fremsat lignende Synspunkter, og jeg haaber derfor paa Bistand fra Regeringens højtærede kvindelige Medlem til at faa en fornuftig Ændring af Lukkeloven, naar den nu forhaabentlig om et halvt Aar kan tages op til en virkelig Revision.

Niels Christian Christensen, Liberal Party

»(...) I therefore hope for support from our honorable female member to enact a reasonable change of the business hours law (...)
«

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Causal inference

- for causal identification, use *politician characteristic regression discontinuity* (PCRD) designs
- → compares narrowly elected candidates with and without a given characteristic in an RD setup
- use method from Luechinger, Schelker & Schmid (2023) to measure closeness in PR elections
- Marshall (2022) critique: PCRD designs do not identify LATE of characteristic X, but of compound of X + 'compensating differentials'
- tentative solution: redefine estimand to include compensating differentials

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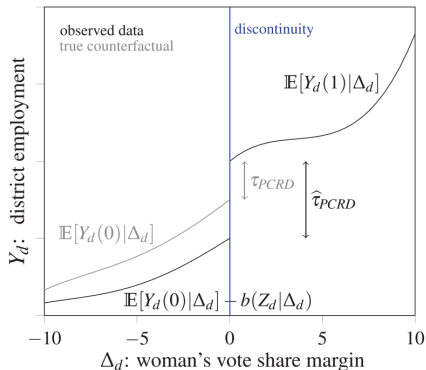
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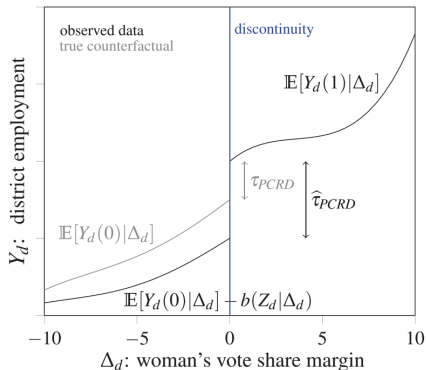


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- funding for two 3-year postdocs, one 1-year full-time RA
- funding for cloud computing etc.
- 3 subprojects:
 - SP1: Digitization of historical election data
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	Subproject 1	Subproject 2	Subproject 3
Responsible	PI + RA	PI + Postdoc 1	PI + Postdoc 2
Focus	Digitization of historical data	Effects of women's candidacies	Effects of workers' candidacies
Advisory board liaison	Tine Paulsen, Univ. of Southern California	Ana Catalano Weeks, Univ. of Bath	Nicholas Carnes, Duke University

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- ways to separate candidate characteristics from compensating differentials?
- other outcomes to consider?

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